Ashutosh Upadhyay

Principal Engineer · Agentic AI, Python & Cloud Data Platforms

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PROFILE

Principal Engineer with 12+ years in enterprise cloud and data engineering, currently building production AI systems for pharmaceutical research and development at Johnson & Johnson Innovative Medicine. Python is my primary engineering language: I use it to build multi-agent LLM applications (Strands Agents SDK, Amazon Bedrock, FastAPI, asyncio), data pipelines for AI (pyarrow, pandas, Parquet, DynamoDB), AWS automation (boto3), observability (OpenTelemetry, Langfuse) and cost analytics. I document what I build, including what failed and how it was fixed, in a public technical blog of 29 long-form posts published in 2026.

EDUCATION

Bachelor of Engineering (B.E.), Electronics & Telecommunication Engineering 2009 – 2013
MGM College of Engineering & Technology, India

PYTHON TECHNICAL SKILLS

Language & standard library	asyncio (`gather`, `run_in_executor`), contextvars, type hints, decorators, classmethod factories, hmac / hashlib, structured logging, json, pathlib, datetime, XML parsing
AI & LLM engineering	Strands Agents SDK (tools, multi-agent orchestration, conversation managers, prompt caching), Amazon Bedrock (Converse / InvokeModel), Amazon Bedrock AgentCore Memory, Langfuse SDK, LLM-as-judge evaluation
Web services & APIs	FastAPI (async handlers, background tasks), requests, REST back ends for SvelteKit front ends
Data engineering	pyarrow (CSV → Parquet, explicit schemas, compute), pandas, Parquet on Amazon S3, Amazon Athena, DynamoDB, SQLite
AWS automation (boto3)	S3, STS AssumeRole (cross-account), CloudWatch & CloudWatch Logs, DynamoDB, Cost Explorer, botocore timeouts and retries
Observability	OpenTelemetry Python SDK (OTLP export, W3C trace propagation), AWS Distro for OpenTelemetry, Prometheus metrics, Langfuse tracing
Imaging, graphics & media	tifffile (OME-TIFF / OME-XML), Pillow, Manim
Running Python in production	Docker, Kubernetes / Amazon EKS (Deployments, Jobs, CronJobs), KEDA, Karpenter, Argo CD, Jenkins
Other languages	SQL, R (Shiny), TypeScript / Svelte, Bash, Terraform

PYTHON PROJECTS & ENGINEERING

> Johnson & Johnson Innovative Medicine, Beerse, Belgium · Principal Engineer · Dec 2025 – present, unless noted otherwise. All systems below run in production on AWS.

1. Multi-agent scientific research platform — Python · Strands Agents SDK · FastAPI · asyncio · Amazon Bedrock · Amazon EKS

- Built and operate a production multi-agent research assistant for drug-discovery scientists: 40 specialised Python tools over 23 scientific data sources, with four agent roles (evidence gathering, analysis, synthesis and an adversarial critic).
- Wrote the asynchronous orchestrator (`asyncio.gather` over Strands `invoke_async`), giving each task a fresh agent instance so concurrent queries never share conversational state.
- Designed the critic to check each synthesis against the raw retrieved data rather than the report alone; it triggers a revision in about 23% of research-tier queries (11 of 47 recorded).
- Handled context-window limits by subclassing a Strands conversation manager to cap tool-result size without dropping messages, recovering partial results on overflow, and sizing sub-agent windows for up to 74 tool calls.
- Stopped async FastAPI handlers from blocking the event loop by moving synchronous Bedrock calls into `run_in_executor` with explicit botocore read timeouts and retry limits.
- Led the upgrade to Claude Opus 5 on Amazon Bedrock: block-type response parsing so reasoning blocks cannot hide the answer, `stop_reason` checks for truncated output, reasoning-effort tuning, a boto3 ≥ 1.40 floor and Strands prompt caching.
- Reordered the system prompt so its static part is a fixed cacheable prefix, raising the prompt-cache hit rate from about 70% to 85%; moved user identity out of the system prompt so it is byte-identical for every user.
- Built a runtime "skills" layer: a Python tool loads curated domain-interpretation rules from S3 as JSON at query time, so domain knowledge can be updated without retraining or redeploying.

2. Agent security, memory & observability — contextvars · boto3 · Bedrock AgentCore Memory · OpenTelemetry · Langfuse

- Found and fixed a cross-tenant access flaw (insecure direct object reference) in which the language model supplied the user identity passed to a tool. Identity now comes from a request-scoped `contextvars.ContextVar` and fails closed, and each user's agent is bound only to the tools that user is entitled to. Fixed during build-out, before any regulated data was connected.
- Integrated cross-session memory with Amazon Bedrock AgentCore Memory through boto3, pseudonymising user IDs with keyed HMAC-SHA256 and routing user feedback into memory with FastAPI background tasks.
- Diagnosed a memory "self-poisoning" loop, where the agent's own mistaken statements were stored and reloaded as facts. Fixed it by persisting turns only after successful tool calls, labelling injected memory with its source, and raising the retrieval threshold from 0.25 to 0.45 based on measured score distributions.
- Instrumented the pipeline with the OpenTelemetry Python SDK (OTLP export, W3C trace context across services) and the Langfuse SDK (tracing, prompt management, LLM-as-judge scores, cost queries). Correcting reasoning-token pricing brought cost estimates to within 2–3% of actual invoices.
- Wrote a boto3/DynamoDB job that uses an LLM to cluster users' written corrections into candidate domain-rule updates for human review.

3. Data engineering for AI — pyarrow · pandas · Parquet · Amazon Athena · DynamoDB · boto3

- Built scheduled Python ingestion pipelines from 20+ scientific data sources into a Parquet data lake on S3 and a DynamoDB knowledge graph. JSON manifests act as commit records, so a freshness check is a single S3 read and the agent reports "not ingested" or "stale" instead of silently returning an empty answer.
- Enforced explicit pyarrow schemas at the ingestion boundary, fixing silent type inference, wrong delimiters on tab-separated files, `large_string` type checks and null propagation in compute filters.
- Led a pre-build design review of a ~129 GB transcriptomics ingestion pipeline that caught five blocking defects; a single ranged S3 request corrected the storage estimate by 3×. The design specifies idempotent writes, resumable checkpoints and explicit Parquet schemas.
- Diagnosed and fixed at source eight "silent wrong data" failures, where results were wrong but no error was raised: for example, case-folding that merged 79 distinct identifiers, XML accessions read from the wrong element, and a DynamoDB counter that inflated knowledge-graph node counts 4×.

4. AI cost analytics (FinOps) & cloud automation — boto3 · pandas · SQLite · CloudWatch · Cost Explorer

- Built the Python data-extraction and analytics back end of an AI cost dashboard tracking Amazon Bedrock, Claude Code and Kiro spend across seven AWS accounts.
- Wrote a cross-account CloudWatch Logs extractor: IRSA plus STS AssumeRole with cached credentials renewed five minutes before expiry, paginated queries over 12-hour windows tracked in SQLite, and region discovery that prevents silent zero-result queries.
- Fixed three silent cost-attribution bugs with pandas: case-normalised identity joins (match coverage from 1.8% to 94.9%), proportional group attribution checked by a conservation invariant (removing 12.6% over-counted credit over 30 days), and UTC calendar-day bucketing.
- Recovered three days of cost data silently lost to a `KeyError` swallowed in a pagination callback, then added a truncated-day detector and load-time pricing validation, and corrected model prices overstated by up to about 89×.
- Built adoption analytics for AI coding assistants, including a pandas fix to how billing caps scale with the reporting window and a mapping from IAM role ARNs to readable team names.

5. LLM evaluation, applied AI & scientific computing

- Built an internal LLM evaluation harness in Python: 33 challenges across 10 R&D domains with about 100 deterministic assertions, an LLM judge, repeated runs and bootstrap 95% confidence intervals, plus a single configuration-driven construction path tested through the real job entry point.
- Delivered a portfolio-modelling AI agent (FastAPI and the Strands Agents SDK, with a SvelteKit interface) that lets analysts query Monte Carlo simulation results in plain language, with guardrails against hallucinated answers.
- Deployed a GPU digital-pathology inference pipeline on Amazon EKS (KEDA event-driven scaling, Karpenter GPU nodes) and fixed its Python OME-TIFF/OME-XML metadata encoding (tifffile) for whole-slide-image overlays.
- Built an animated walkthrough of a multi-agent query in Manim (Python), narrated with Amazon Polly, for my internal talk "When one AI isn't enough"; wrote the Pillow scripts that generate my blog's cover images.
- 2020 – 2024: Built a platform for hosting Python and R Shiny applications and Streamlit apps on Amazon EKS with ShinyProxy and other open-source tools.

PROFESSIONAL EXPERIENCE

Johnson & Johnson Innovative Medicine — Principal Engineer — Dec 2025 – present

Beerse, Belgium

- Lead engineer for the agentic AI platforms and Python systems described above (projects 1–5).
- Set up Kargo on Amazon EKS with Microsoft Entra ID single sign-on and Argo CD integration. Designed a promotion pipeline from development to production for research workloads, with a manual approval gate before production.

- Drive adoption of AI-assisted engineering (Claude Code, Kiro) across platform and data teams.

Amazon Web Services (AWS) — Solutions Architect (Partner) Dec 2024 – Dec 2025
Brussels, Belgium

- Technical lead on partner implementation projects: defined architectures, responded to requests for proposals, and led in-depth architecture reviews before production deployment.

Johnson & Johnson Innovative Medicine — Principal Engineer · Lead Engineer · Senior IT Analyst Sep 2020 – Dec 2024
Beerse, Belgium

- Led the build of AWS platforms that scientists and biologists use for data analytics and modelling. This included a proteomics data-analysis pipeline on Amazon EKS for the Bioinformatics team.
- Delivered cost optimisation across 12 AWS accounts, with annual savings of over $1.5M. Partnered with AWS on testing the open-source Karpenter autoscaler, saving over $500K a year on EKS workloads.
- Built a hosting platform for Python and R Shiny applications and Streamlit apps on Kubernetes, using ShinyProxy. Set up GitOps with Argo CD, Argo Rollouts and Argo Events.
- Led an ETL migration of high-performance computing (HPC) data from MongoDB to Amazon RDS for PostgreSQL, feeding Power BI reporting.
- Selected for the AWS Subject Matter Expert programme to write exam items for the AWS Solutions Architect – Professional and DevOps Engineer – Professional certifications.

LTI (Larsen & Toubro Infotech) — Senior Software Engineer Oct 2017 – Sep 2020
Navi Mumbai, India → Beerse, Belgium

- Worked as an AWS technical specialist embedded at Johnson & Johnson for the EMEA region, covering Redshift, EMR, Athena and Talend ETL. Led the migration of the entire AWS production environment to a new account.

Earlier roles Apr 2014 – Oct 2017

- **Amdocs**, Business Intelligence Developer (2017): built data-virtualisation views across Teradata, Oracle, Redshift, Vertica and HDFS.
- **BlazeClan Technologies**, Cloud Engineer (2016–2017): AWS architecture and data profiling with Talend, Redshift and Tableau.
- **Capgemini**, Software Engineer → Senior Software Engineer (2014–2016): ETL pipelines in DataStage and Teradata. Automated a manual job, cutting its runtime from 45 hours to 30 minutes.

TECHNICAL WRITING

Field Notes (aupadh12.github.io/enigmaAI): 29 long-form posts (2026) on building and running production AI systems. Selected Python-heavy posts:

- Multi-Agent AI Collaboration in Production
- Your Agent Is a Confused Deputy (agent security)
- The Context-Window Overflow Was Not Where We Looked
- Silent Wrong Data in Research Pipelines
- Upgrading an Agent Platform to Claude Opus 5
- Langfuse in Production: LLM Observability
- Multi-Tenant FinOps: Three Silent Bugs
- Building an Internal LLM Evaluation Suite
- Data Infrastructure Quality for AI Readiness
- IRSA + Cross-Account AssumeRole on EKS

CERTIFICATIONS

AWS: Solutions Architect – Professional · DevOps Engineer – Professional · Generative AI Developer – Professional · Machine Learning – Specialty · Security – Specialty · Data Analytics – Specialty · Database – Specialty · Advanced Networking – Specialty · Machine Learning Engineer – Associate · Data Engineer – Associate · Solutions Architect – Associate · SysOps Administrator – Associate · Developer – Associate · AI Practitioner · Cloud Practitioner. AWS Certification Subject Matter Expert (Foundational, Associate and Professional).

Linux Foundation / CNCF: Certified Kubernetes Application Developer (CKAD) · Certified Argo Project Associate (CAPA) · Kyverno Certified Associate (KCA).

Python & data courses: Data Engineering Nanodegree (Udacity) · Shiny for Python · Building Production-Quality Shiny Applications.

Other: HashiCorp Certified: Terraform Associate · SnowPro Core (Snowflake) · Gremlin Certified Chaos Engineering Practitioner · Tableau Author.

AWARDS & LANGUAGES

Awards: Johnson & Johnson Innovative Medicine Inspire Awards (Bravo, Spotlight, Celebrate) · Technical Achievement Award · Pat on the Back Award (twice).

English: C2 Proficient, EF SET English Certificate (79/100).